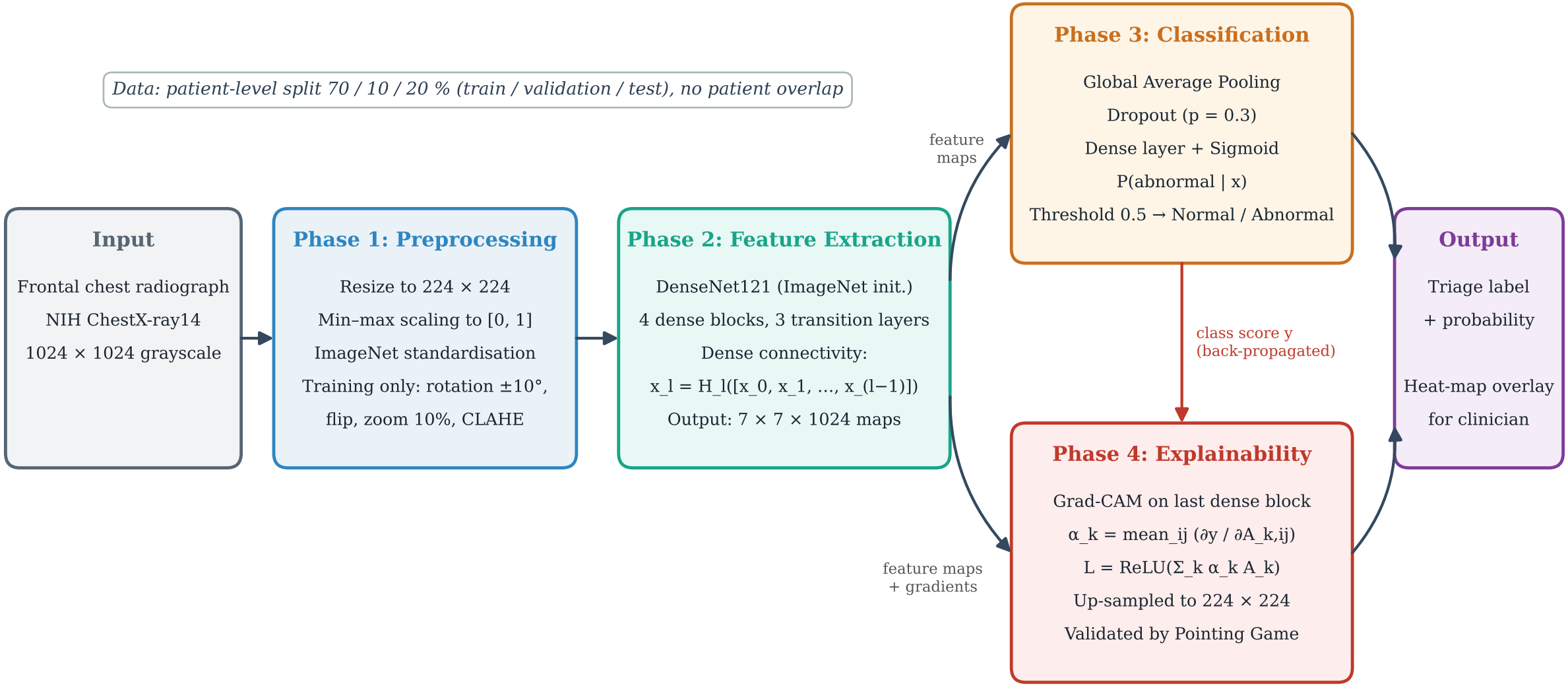
Data: patient-level split 70 / 10 / 20 % (train / validation / test), no patient overlap
Input
Frontal chest radiograph
NIH ChestX-ray14
1024 × 1024 grayscale
Phase 1: Preprocessing
Resize to 224 × 224
Min–max scaling to [0, 1]
ImageNet standardisation
Training only: rotation ±10°,
flip, zoom 10%, CLAHE
Phase 2: Feature Extraction
DenseNet121 (ImageNet init.)
4 dense blocks, 3 transition layers
Dense connectivity:
x_l = H_l([x_0, x_1, ..., x_(l−1)])
Output: 7 × 7 × 1024 maps
feature maps
Phase 3: Classification
Global Average Pooling
Dropout (p = 0.3)
Dense layer + Sigmoid
P(abnormal | x)
Threshold 0.5 → Normal / Abnormal
class score y
(back-propagated)
feature maps
+ gradients
Phase 4: Explainability
Grad-CAM on last dense block
α_k = mean_ij (∂y / ∂A_k,ij)
L = ReLU(Σ_k α_k A_k)
Up-sampled to 224 × 224
Validated by Pointing Game
Output
Triage label
+ probability
Heat-map overlay
for clinician